Second_lecture_subsetting

Vector subsetting by position

Subsetting means that you extract some elements from the vector

```
people <- c("Peter", "Jane", "Tom", "Paul")
weights <- c(90, 65, 67, 74)
people[1]
```

```
[1] "Peter"
```

```
people[1:3]
```

```
[1] "Peter" "Jane"  "Tom"
```

```
people[c(1,3)]
```

```
[1] "Peter" "Tom"
```

```
people[-c(2:4)]
```

```
[1] "Peter"
```

Vector subsetting by a condition

```
weights[weights >= 67]
```

```
[1] 90 67 74
```

Look how the condition actually reads. What is it? Save it in a variable and inspect its class.

```
weights >= 67
```

```
[1]  TRUE FALSE  TRUE  TRUE
```

```
test_variable <- weights >= 67
class(test_variable)
```

```
[1] "logical"
```

So you actually give weights a logical vector, and R picks just the positions, where the element is TRUE:

```
weights[c(TRUE, FALSE, TRUE, TRUE)]
```

```
[1] 90 67 74
```

There is a function that gives you the positions where the condition is TRUE

```
which(weights >= 67)
```

```
[1] 1 3 4
```

Subsetting a data frame

When two vectors have equal length, we can combine them into a data frame. We will use the base R built-in function `data.frame`.

Look at the table: four rows, two columns. In square brackets: two positions, first concerns rows, second concerns columns.

```r
our_df <- data.frame(people, weights)
our_df
```

```
  people weights
1  Peter      90
2   Jane      65
3    Tom      67
4   Paul      74
```

Give me just the first row:

```r
our_df[1, ]
```

```
  people weights
1  Peter      90
```

Give me just the first column - several ways to do it

```r
our_df[1]
```

```
  people
1  Peter
2   Jane
3    Tom
4   Paul
```

or like this

```r
our_df["people"]
```

```
  people
1  Peter
2   Jane
3    Tom
4   Paul
```

or like this

```
our_df[, 1]
```

```
[1] "Peter" "Jane"  "Tom"   "Paul"
```

or like this

```
our_df$people
```

```
[1] "Peter" "Jane"  "Tom"   "Paul"
```

Give me some rows and only the second column

```
our_df[c(1,4), 2]
```

```
[1] 90 74
```